

NORTHWIND GLOBAL SALES & OPERATIONS ANALYTICS

Project Summary

Executive Overview

I developed an interactive **Business Intelligence and Executive Decision-Support Dashboard for Northwind Traders** using Power BI. The project involved transforming transactional sales, customer, product, employee, shipping, and order data into a structured analytical solution that enables management to monitor commercial performance, product health, regional revenue, logistics efficiency, salesforce productivity, and discount strategy.

The project was executed through a complete BI workflow:

Raw Data → Data Preparation → Data Transformation → Data Modeling → DAX Measures → Validation → Interactive Visualization → Business Analysis → Executive Reporting.

The final solution consists of a **three-page Power BI executive dashboard** covering:

1. **Sales & Revenue Overview**
2. **Product & Category Intelligence**
3. **Regional, Operations & People Performance**

In addition to answering the five questions specified in the challenge, I expanded the analysis to answer additional business questions around customer markets, product concentration, discontinued products, freight burden, shipping performance, employee productivity, and discount exposure.

1. Project Objective

The primary objective was to give Northwind Traders' executive team a clear and interactive view of the company's global sales and operational performance.

The dashboard was designed to answer questions such as:

- How is revenue performing over time?

- How many orders are being generated?
- Are there identifiable seasonal patterns?
- Which product categories drive revenue?
- Which individual products contribute the most?
- What is the impact of discontinued products?
- Which countries and cities contribute the most revenue?
- Where are freight costs disproportionately high?
- Which shipping companies provide the best operational performance?
- Which employees generate the highest sales?
- How does discounting vary across product categories?
- Where is the business experiencing potential pricing or promotional pressure?

The solution was therefore designed not merely as a collection of charts, but as an **executive analytical system for decision-making**.

2. Dataset and Data Sources

The analysis was built using the **Northwind Traders dataset**, which contains interconnected information about customers, products, categories, orders, order details, employees, and shippers.

The main datasets used were:

Customers

Contains:

- Customer ID
- Company Name
- Contact Name
- Contact Title
- City
- Country

Products

Contains:

- Product ID
- Product Name
- Quantity Per Unit
- Unit Price

- Discontinued Status
- Category ID

Categories

Contains:

- Category ID
- Category Name
- Description

Orders

Contains:

- Order ID
- Customer ID
- Employee ID
- Order Date
- Required Date
- Shipped Date
- Shipper ID
- Freight

Order Details

Contains:

- Order ID
- Product ID
- Unit Price
- Quantity
- Discount

Employees

Contains:

- Employee ID
- Employee Name
- Title
- City
- Country
- Reports-To

Shippers

Contains:

- Shipper ID
- Company Name

The dataset contains approximately:

Data Component	Records
Orders	830
Order Details	2,155
Products	77
Customers	91
Categories	8
Employees	9
Shippers	3

The order data covers approximately **July 2013 to May 2015**.

3. Technical Architecture

Tools Used

The project was executed using:

- **Power BI Desktop**
- **Power Query**
- **DAX**
- Northwind transactional datasets

Power Query was used for data preparation and transformation, while DAX was used to create reusable business measures and KPIs.

The final analytical layer was presented through a three-page interactive Power BI dashboard.

4. Data Preparation and Transformation

I first imported the seven major datasets into Power BI and used Power Query to prepare them for analysis.

The imported datasets were:

customers
products
categories
orders
order_details
employees
shippers

Data Type Standardization

I reviewed and standardized the data types to ensure calculations and relationships worked correctly.

Examples include:

Customer ID → Text
Product ID → Whole Number
Order ID → Whole Number
Employee ID → Whole Number
Category ID → Whole Number
Shipper ID → Whole Number

Dates were standardized as:

Order Date → Date
Required Date → Date
Shipped Date → Date

Numerical fields were appropriately configured:

Unit Price → Decimal
Quantity → Whole Number
Discount → Decimal

Freight → Decimal

This ensured that Power BI could correctly perform aggregations, date calculations, filtering, and DAX calculations.

5. Product Status Transformation

The original **Discontinued** field used technical values:

0
1

I transformed this into a more understandable business field:

Product Status

0 → Active
1 → Discontinued

This made the dashboard easier for a non-technical executive audience to interpret.

It also enabled direct analysis of the commercial significance of discontinued products.

6. Shipping Data Preparation

Shipping performance was one of the major areas of the project.

I retained orders that had no shipment date instead of deleting them because these records still represent legitimate orders.

I created a calculated **Delivery Days** field based on:

Shipped Date - Order Date

For orders without a shipment date, the delivery duration remains blank.

I also created a **Shipping Status** classification:

No Shipped Date
→ Not Shipped

Shipped Date $\leq$ Required Date
→ On Time

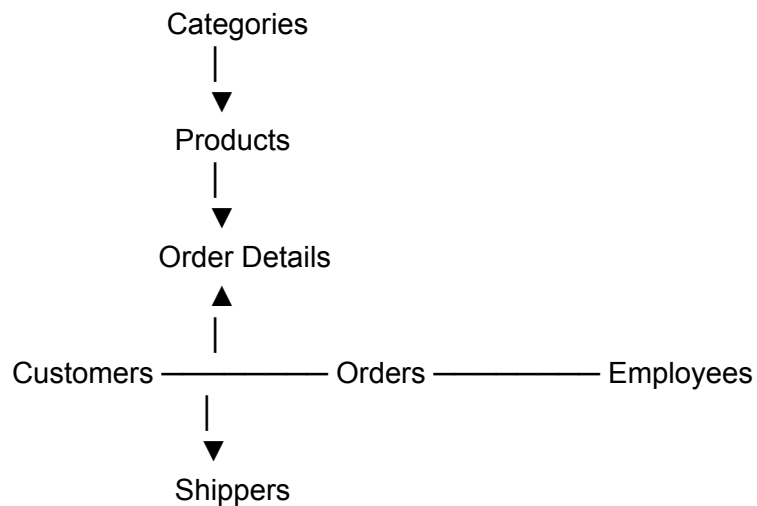
Shipped Date $>$ Required Date
→ Late

This allowed shipping performance to be analyzed without incorrectly treating unshipped orders as late shipments.

7. Data Model

I established relationships between the tables so that information could flow correctly across the analytical model.

The model follows a relational structure:



Key relationships include:

Customers 1 → Many Orders

Employees 1 → Many Orders

Shippers 1 → Many Orders

Orders 1 → Many Order Details

Products 1 → Many Order Details

Categories 1 → Many Products

This model allowed filters such as **Year, Country, Category, Employee, Product, and Customer** to interact with the relevant analytical visuals.

8. Date Dimension

I created a dedicated **Date Table** to support proper time-series analysis.

The Date Table contains fields including:

Date
Year
Month Number
Month
Month Short
Quarter
Year Month
Year Quarter

The Date Table was connected to the order date and configured as the model's date dimension.

I also sorted the month names using the month number to prevent Power BI from displaying months alphabetically.

This enabled:

- Monthly analysis
 - Quarterly analysis
 - Yearly analysis
 - Revenue trends
 - Order trends
 - Seasonal analysis
-

9. DAX Measures

I created a dedicated **Measures** table to organize the analytical calculations.

The major measures include:

Gross Sales

Calculates sales before discounts:

Unit Price $\times$ Quantity

Discount Amount

Calculates the value given away through discounts:

Unit Price $\times$ Quantity $\times$ Discount

Revenue

Calculates revenue after discounts:

Gross Sales - Discount Amount

Total Orders

Counts unique orders.

Units Sold

Calculates total product quantity sold.

Average Order Value

Revenue $\div$ Total Orders

Total Freight

Calculates total freight expenditure.

Average Freight

Calculates average freight per order.

Discount Rate

Discount Amount $\div$ Gross Sales

Average Delivery Days

Calculates the average delivery duration across completed shipments.

On-Time Orders

Counts orders shipped on or before the required date.

Late Orders

Counts orders shipped after the required date.

Not Shipped Orders

Counts orders without a shipment date.

Shipped Orders

On-Time Orders + Late Orders

On-Time Rate

On-Time Orders ÷ Shipped Orders

This excludes unshipped orders from the denominator.

Freight % of Revenue

Total Freight ÷ Revenue

Revenue per Unit

Revenue ÷ Units Sold

These measures provided the foundation for the dashboard's KPIs and analytical visuals.

10. Model Validation

Before finalizing the dashboard, I validated the model and measures.

I checked that:

- Total Orders returned the expected **830 orders**
- Revenue calculations produced valid results
- Country filters changed relevant metrics
- Category filters changed relevant metrics

- Employee filters affected sales metrics
- Product filters affected product metrics
- Relationships propagated filters correctly
- Shipping calculations excluded unshipped orders from delivery-time analysis

This validation stage was important because a visually attractive dashboard is not useful if the underlying calculations or relationships are incorrect.

11. Dashboard Design

The final dashboard contains three analytical pages.

The design was intentionally kept clean and executive-focused rather than overcrowding the pages with unnecessary visuals.

The dashboard uses a consistent visual hierarchy, card-based KPI presentation, aligned charts, controlled colors, clear titles, and interactive filters.

12. Page 1 — Sales & Revenue Overview

Executive Purpose

The first page establishes the overall commercial health of Northwind Traders.

It answers:

How is Northwind performing overall, and how has sales performance changed over time?

Key KPIs

The page presents:

- **Revenue:** \$1.27M
- **Total Orders:** 830
- **Units Sold:** 51.3K
- **Average Order Value:** \$1,525
- **Total Freight:** \$64.9K

More specifically:

Revenue = \$1,265,793.04

Orders = 830

Units Sold = 51,317

AOV = \$1,525.05

Freight = \$64,942.69

Main Visuals

Revenue Trend Over Time

A monthly line chart showing revenue progression across the available period.

Order Volume Trend

A monthly line chart showing transaction volume.

Top Markets by Revenue

A horizontal bar chart showing the countries contributing the most revenue.

Executive Summary

A narrative section was included to help communicate the most important findings to management.

13. Page 2 — Product & Category Intelligence

Executive Purpose

The second page investigates the health of Northwind's product portfolio.

It answers:

Which products and categories are driving revenue, and where are there potential portfolio and discount risks?

KPIs

Revenue = \$1.27M

Units Sold = 51.3K

Discount Amount = \$88.7K

Discount Rate = 6.55%

Revenue by Category

Beverages was the strongest category at approximately:

\$267.9K

followed by:

Dairy Products — \$234.5K

Together, Beverages and Dairy Products account for approximately **39.7% of total revenue**.

Units Sold by Category

Beverages also lead in unit volume with approximately:

9.5K units

followed by Dairy Products at approximately:

9.1K units

Top 10 Products by Revenue

The analysis identified **Côte de Blaye** as the strongest individual product:

\$141.4K

followed by:

Thüringer Rostbratwurst — \$80.4K

Product Portfolio Status

The product portfolio contains:

69 Active Products

8 Discontinued Products

Therefore:

Active = 89.6%

Discontinued = 10.4%

Revenue by Product Status

An important discovery was that discontinued products still generated substantial historical revenue:

Active Products

\$1.08M / 85.4%

Discontinued Products

\$185.0K / 14.6%

This means only **10.4% of the product catalog is discontinued**, but those products historically generated **14.6% of revenue**.

That creates a product portfolio risk and replacement opportunity.

Discount Analysis

Total discounts amounted to approximately:

\$88,665.55

with an overall discount rate of:

6.55%

Beverages had the largest discount amount:

\$18.7K

while Dairy Products followed at approximately:

\$16.8K

However, **Meat & Poultry** had the highest discount rate:

8.51%

This indicates that the category experiences relatively higher promotional pressure.

Product Detail Table

A detailed table was also included showing product-level information such as:

- Product
 - Category
 - Product Status
 - Unit Price
 - Units Sold
 - Revenue
 - Discount Amount
-

14. Page 3 — Regional, Operations & People Performance

Executive Purpose

The third page brings together geographical, logistics, and employee performance.

It answers:

Where is revenue coming from, where are logistics costs creating friction, which shipping providers perform best, and which employees are driving sales?

KPIs

Revenue = \$1.27M

Total Freight = \$64.9K

Average Delivery Days = 8.49

On-Time Delivery Rate = 95.43%

Regional Performance

The dashboard analyzes revenue by:

- Country
- City
- Market

It also calculates freight burden as a percentage of revenue.

Freight Burden

Overall freight represents approximately:

5.13% of revenue.

Some markets experience significantly higher freight pressure.

The highest examples include:

Argentina → 7.37%

Sweden → 5.92%

Austria → 5.80%

These markets may require further investigation into shipping routes, carrier selection, order consolidation, or pricing strategy.

Top Cities by Revenue

The strongest cities include:

Cunewalde → ~110K *Graz* → ~105K

Boise → ~\$104K

15. Shipping / Shipper Performance

The dashboard compares the three shipping providers across:

- Total Orders
- Freight Spend

- Average Freight per Order
- Delivery Days
- On-Time Rate

Federal Shipping

Average Delivery = 7.47 days

On-Time Rate = 96.39%

Average Freight = \$81.78/order

Federal Shipping provides the strongest combination of speed and reliability.

Speedy Express

Average Freight = \$65.45/order

Delivery = 8.57 days

On-Time Rate = 95.10%

Speedy Express offers the lowest shipping cost per order while maintaining a relatively strong service level.

United Package

Orders = 315

Average Freight = \$87.48/order

Delivery = 9.23 days

On-Time Rate = 94.92%

United Package handles significant volume but has the:

- Highest average freight cost
- Longest delivery time
- Lowest on-time rate

This makes it an important candidate for operational review.

16. Salesforce Performance

The dashboard also evaluates employees based on:

- Revenue
- Number of orders
- Average order value

Revenue Leaders

The leading employees include:

Margaret Peacock

Revenue = \$232.9K

Orders = 156

Janet Leverling

Revenue = \$202.8K

Orders = 127

Nancy Davolio

Revenue = \$192.1K

High Average Deal Size

The analysis also revealed that high revenue does not always mean high order volume.

For example:

Anne Dodsworth

43 Orders

77.3K Revenue ~ 1,798 AOV

Andrew Fuller

96 Orders

166.5K Revenue ~ 1,735 AOV

This distinction is important because it separates **high-volume salespeople** from **high-value-deal salespeople**.

Volume vs Revenue Analysis

A scatter plot was used to compare:

Order Volume

VS

Revenue

This provides a more sophisticated view of salesforce productivity than ranking employees using revenue alone.

17. Dashboard Interactivity

The dashboard includes interactive filters for:

- **Year**
- **Country**
- **Category**

These filters allow management to move from a high-level view to more specific analysis.

For example:

What was revenue in 2014?

Which category performed best in a particular country?

How does freight performance change by market?

Which products are responsible for category revenue?

The slicers were synchronized across the three dashboard pages.

A **Reset Filters** button was also implemented using Power BI bookmarks so that users can quickly return to the default dashboard view.

Visual interactions were configured to provide appropriate filtering rather than leaving Power BI's default highlighting behavior everywhere.

18. Visual Design System

I used a consistent executive/business intelligence visual design throughout the dashboard.

Primary Color Palette

Primary Navy

#17324D

Used for:

- Page titles
- Visual headers
- Primary revenue visuals
- Main dashboard elements

Secondary Blue

#2F6B9A

Used for:

- Volume charts
- Trend visuals
- Secondary comparisons
- Data bars

Operational Green

#2E8B57

Used for:

- Positive operational performance
- On-time performance
- Favorable indicators

Warning Orange

#F4A261

Used for:

- Freight pressure
- Discount amounts
- Operational attention areas

Alert Red

#D9534F

Used for:

- Higher discount rates

- Risk indicators
- Negative performance indicators

Background

#F5F7FA

Containers

White:

#FFFFFF

with subtle borders:

#E2E8F0

19. Typography and Layout

The dashboard uses a clean hierarchy based on **Segoe UI**.

The formatting hierarchy includes approximately:

Page Titles → 24px Bold

Visual Titles → 13px Bold

KPI Values → 24px Bold

Supporting Values → 9.5–10px

The dashboard uses:

- Consistent card dimensions
- Proper spacing
- Alignment
- White visual containers
- Subtle borders
- Minimal visual clutter
- Clear chart titles
- Consistent currency formatting
- Percentage formatting

- Whole-number formatting

The goal was to make the dashboard look like an **executive reporting product**, rather than a collection of unrelated Power BI visuals.

20. Key Business Discoveries

The analysis produced several important findings.

1. Strong Revenue Growth and Seasonality

Annual revenue increased from approximately:

\$208.1K in 2013

to:

\$617.1K in 2014.

The data also shows recurring increases in sales activity during the **October–December period**, suggesting a potential Q4 seasonal pattern.

The highest monthly revenue observed was approximately:

\$123.8K in April 2015.

2. Revenue Is Concentrated in a Few Product Categories

Beverages and Dairy Products are the strongest categories.

Combined, they contribute approximately:

39.7% of total revenue.

This indicates significant dependence on a relatively small number of product categories.

3. Discontinued Products Represent a Significant Revenue Opportunity

Only:

8 of 77 products

are discontinued.

That represents approximately:

10.4% of the product catalog.

However, those discontinued products historically generated:

\$184,990.93

or approximately:

14.61% of total revenue.

This is an important strategic finding.

Some discontinued products are not insignificant; they include:

- Thüringer Rostbratwurst
- Alice Mutton

Therefore, product discontinuation decisions should be evaluated alongside historical sales contribution, and suitable replacement products should be identified.

21. Discount Strategy Findings

The business provided approximately:

\$88,665.55 in discounts

representing an overall discount rate of:

6.55%.

Beverages received the largest absolute discount amount.

However, Meat & Poultry recorded the highest proportional discount rate:

8.51%.

This suggests that discount strategy should not be evaluated solely by total discount value.

Management should also consider:

- Discount rate
- Revenue generated
- Order volume
- Product category
- Customer behavior

to determine whether promotional spending is generating sufficient commercial value.

22. Logistics Findings

Overall freight represents approximately:

5.13% of revenue.

However, freight pressure differs substantially across markets.

Argentina, Sweden, and Austria showed relatively high freight-to-revenue ratios.

This indicates that logistics efficiency is not uniform across Northwind's markets.

Management could investigate:

- Carrier allocation
 - Shipping routes
 - Order consolidation
 - Minimum order quantities
 - Market-specific pricing
 - Freight recovery strategies
-

23. Salesforce Findings

The analysis demonstrated that salesforce productivity should not be measured by revenue alone.

Margaret Peacock is a strong **volume and revenue driver**, while employees such as Anne Dodsworth and Andrew Fuller demonstrate relatively high average order values.

This means Northwind can segment its salesforce into categories such as:

High Volume / High Revenue

High Value / Lower Volume

High Volume / Lower AOV

Lower Volume / Lower Revenue

This provides a stronger basis for coaching, target setting, incentive design, and resource allocation.

24. Analytical Boundaries and Limitations

I deliberately documented the limitations of the dataset instead of making unsupported assumptions.

Limitation 1 — No COGS

The dataset does not contain:

- Cost of Goods Sold
- Product acquisition cost
- Manufacturing cost
- Operating costs

Therefore, true:

- Profit
- Gross Profit
- Gross Margin
- Net Profit Margin

cannot be calculated reliably.

Instead, I evaluated the impact of discounting using:

- Gross Sales
- Discount Amount
- Discount Rate
- Revenue

This prevents unsupported profitability claims.

Limitation 2 — Market Share

The dataset only contains Northwind's own sales.

It does not contain the total external market size for each country or city.

Therefore, the regional analysis represents:

Northwind Revenue Share

rather than true external market share.

For example, it would be incorrect to say:

"Northwind owns 20% of the German market."

The defensible statement would be:

"Germany contributes X% of Northwind's recorded revenue."

Limitation 3 — Missing Shipping Dates

There are exactly:

21 orders

without a `shippedDate`.

Therefore:

830 Total Orders

-

21 Not Shipped

=

809 Shipped Orders

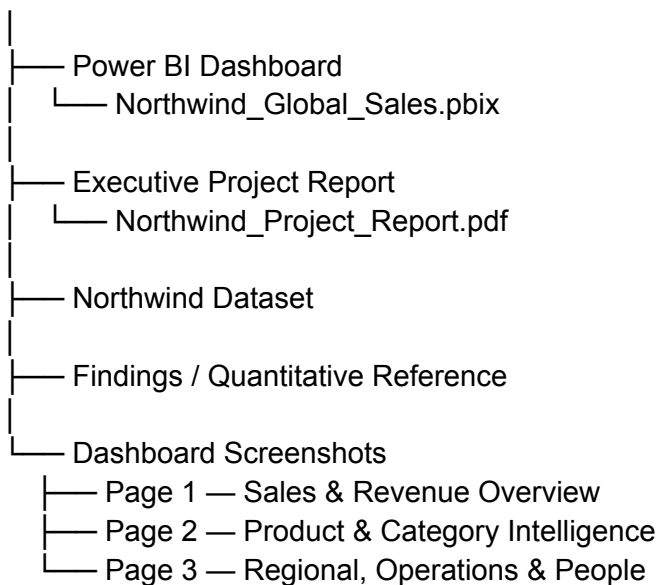
Delivery time and on-time performance were therefore evaluated across the **809 completed shipments**.

The 21 unshipped orders were classified separately as **Not Shipped** rather than being treated as late.

25. Final Deliverables

The completed Task A project consists of the following components:

TASK A — NORTHWIND GLOBAL SALES & OPERATIONS



The project report is intended to contain the **shareable Power BI dashboard link**, as required by the challenge.

The completed files should ultimately be organized inside the required **Google Drive submission folder**.

26. Complete Project Workflow

The entire work I executed can be represented as:

NORTHWIND RAW DATA



DATA IMPORT



POWER QUERY



- Data Type Cleaning
- Product Status Transformation
- Shipping Calculations
- Data Preparation



DATA MODEL



- Customers
- Orders
- Order Details
- Products
- Categories
- Employees
- Shippers



DATE DIMENSION



DAX MEASURES

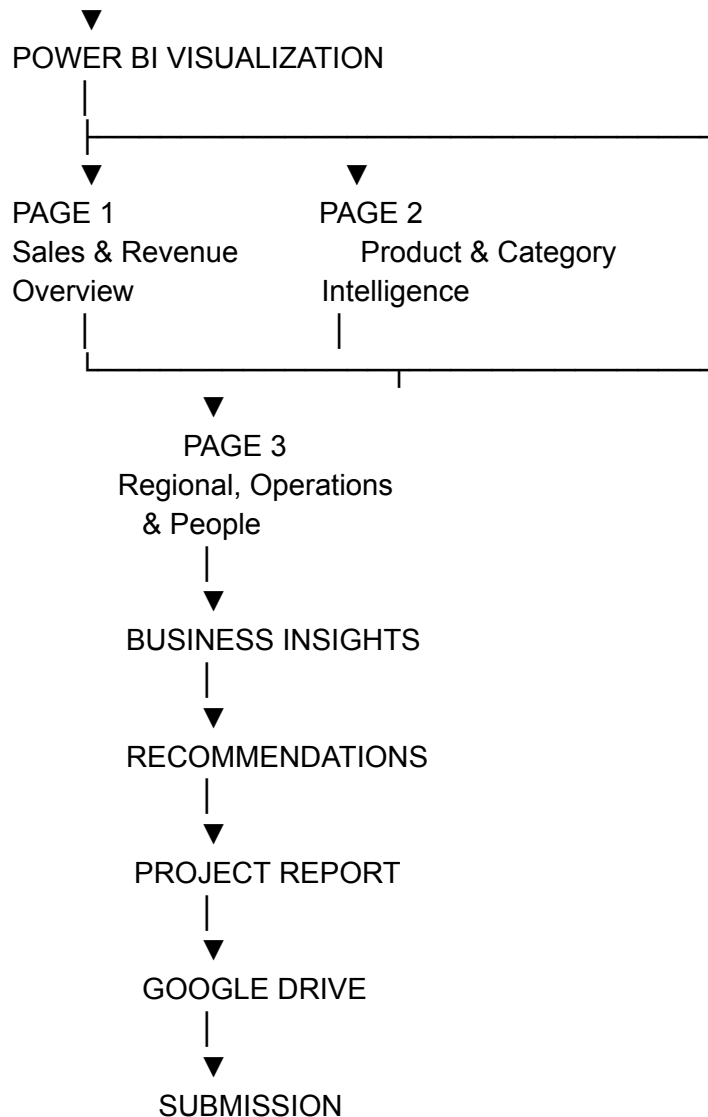


- Revenue
- Orders
- Units
- AOV
- Freight
- Discounts
- Delivery
- On-Time Performance



MODEL VALIDATION





27. Overall Outcome

The completed Northwind project demonstrates my ability to take a **raw transactional dataset** and turn it into an **end-to-end Business Intelligence solution**.

The project goes beyond simply creating charts. It demonstrates my ability to:

- Understand a business problem
- Profile and prepare raw data
- Build a relational data model
- Create analytical DAX measures
- Validate calculations
- Design an executive dashboard

- Build interactive reporting
- Identify commercial trends
- Analyze product performance
- Evaluate operational efficiency
- Analyze Salesforce productivity
- Identify data limitations
- Translate analytical findings into business recommendations

The most important outcome is that the dashboard allows an executive to move from:

"What happened?"

to:

"Where did it happen?"

to:

"Why might it be happening?"

and ultimately:

"Where should management focus?"

That is the core value of the Northwind BI solution.

One-line portfolio description

Built an end-to-end Power BI Business Intelligence solution for Northwind Traders, transforming 830 orders and 2,155 order-line records into a three-page interactive executive dashboard covering revenue trends, product portfolio performance, discount exposure, regional markets, logistics efficiency, and Salesforce productivity.